

BLUESTOCK MUTUAL FUND DATA ANALYTICS REPORT

Capstone Project I: Advanced ETL Architecture, Quantitative Performance
Risk Modeling, and Interactive Web Dashboard Delivery

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Executive Summary

In modern financial markets, the proliferation of retail mutual fund investments demands robust quantitative risk models and highly interactive decision-support tools. This capstone project delivers an end-to-end data engineering and business intelligence solution for Bluestock Fintech. By unifying disjointed raw data streams, constructing a star-schema relational database, executing advanced quantitative performance analytics, and building a high-fidelity interactive dashboard, we solve the challenge of mutual fund portfolio evaluation.

Core ETL Architecture: We built a python-based ETL pipeline that cleans raw NAV histories and transaction logs, standardizes categories, handles calendar alignments, and ingests them into a structured SQLite database. Optimization via indexing ensures sub-second query execution times across over 40,000 NAV records.

Advanced Financial Modeling: The quantitative analytics engine computes compound annualized growth rates (CAGRs), Sharpe/Sortino ratios, market risk regressions (Alpha, Beta), and worst drawdowns. We also modeled tail-risk metrics (Historical VaR and Conditional VaR at 95% confidence) and calculated Sector Herfindahl-Hirschman Indices (HHI) to evaluate portfolio concentration risk.

Dashboard & Business Insights: A premium web dashboard was developed implementing Bluestock's branding. Key behavioral insights indicate a 28% increase in retail SIP ticket sizes in the 2025 cohort, but highlight a 28.45% customer retention leakage due to skipped monthly SIP contributions. Strategic recommendations focus on retention campaign implementation and sector-specific asset rebalancing.

Data Sources & Data Dictionary

The project integrates ten distinct processed datasets structured in a centralized SQLite database. Below is the metadata description and schema specification for the core dimension and fact tables:

Table Name	Type	Key Columns	Description
dim_fund	Dimension	amfi_code (PK), fund_house, category, plan, risk_category, expense_ratio	Metadata for 40 mutual fund schemes.
dim_date	Dimension	date (PK), year, month, quarter, is_weekend, day_name	Pre-computed calendar dimensions (2022-2026).
fact_nav	Fact	nav_id (PK), amfi_code (FK), date (FK), nav	Daily NAV historical records (40,000+ rows).
fact_transactions	Fact	transaction_id (PK), investor_id, amfi_code (FK), amount_inr, type	Investor transaction logs (SIP/Lumpsum/Redemption).
fact_performance	Fact	amfi_code (PK, FK), sharpe_ratio, risk_grade, return_3yr_pct, aum_crore	Pre-calculated fund quantitative performance stats.
fact_holdings	Fact	holding_id (PK), amfi_code (FK), sector, weight_pct	Individual stock holdings and sector allocations.
fact_aum	Fact	aum_id (PK), date (FK), fund_house, aum_crore, num_schemes	Historical assets under management snapshots per AMC.

Table Schemas Details (dim_fund & fact_transactions):

Column Name	Data Type	Constraints	Definition
amfi_code	INTEGER	PRIMARY KEY	Unique identifier assigned to each scheme by AMFI.
fund_house	TEXT	NOT NULL	Asset Management Company (AMC) name.
scheme_name	TEXT	NOT NULL	Full mutual fund scheme name.
category	TEXT	NOT NULL	Asset class category (Equity, Debt, Hybrid).
plan	TEXT	CHECK(plan IN ('Direct','Regular'))	Direct vs. Regular commission structure.
expense_ratio_pct	REAL	CHECK(>= 0.0)	Annualized fund management fee percentage.

Data Dictionary (Continued)

Table: fact_transactions Schema

Column Name	Data Type	Constraints	Definition
transaction_id	INTEGER	PRIMARY KEY AUTOINCREMENT	Unique serial transaction ID.
investor_id	TEXT	NOT NULL	Masked investor alphanumeric ID.
transaction_date	TEXT	FOREIGN KEY (dim_date)	Date of transaction (YYYY-MM-DD).
amfi_code	INTEGER	FOREIGN KEY (dim_fund)	Target mutual fund code.
transaction_type	TEXT	CHECK IN ('SIP','Lumpsum','Redemption')	Directional flow of transaction.
amount_inr	REAL	CHECK(amount_inr > 0)	Monetary size of transaction in INR.
state	TEXT	NOT NULL	Indian state of investor residence.
city_tier	TEXT	CHECK IN ('T30', 'B30')	City tier classification (Top 30 vs. Beyond 30).
age_group	TEXT	-	Investor age group bracket.
gender	TEXT	-	Investor gender (Male/Female).

ETL Pipeline & Relational Star Schema Design

The ETL architecture was constructed to handle multi-source ingestion and maintain high data integrity. The pipeline comprises three distinct phases: Extraction, Transformation, and Ingestive Loading.

1. Data Extraction & Ingestion:

Raw datasets were loaded from tabular flat files (`data/raw/`). A secondary live execution loader was built in Python (`scripts/live\_nav\_fetch.py`) to simulate dynamic AMFI API integration, enabling continuous NAV stream fetches. All files are ingested via pandas dataframes before processing.

2. Data Transformation & Cleaning:

Data transformation was structured to enforce quality control rules. Core cleaning operations in `scripts/data\_cleaning.py` include:

- **Calendar Alignment:** NAV records are missing on market holidays and weekends. To resolve this, a daily calendar from Jan 2022 to May 2026 was generated, and missing values were forward-filled from the previous business day (`ffill()`).
- **Anomalous Filter:** Flagged and cleaned invalid negative NAV inputs, expense ratio anomalies outside the `[0.1%, 2.5%]` interval, and standardized category fields.

3. Ingestive Database Loader:

The SQLite loader (`scripts/db\_loader.py`) maps the transformed tables into the star-schema database. SQLite parameters enforce strict integrity rules: foreign key constraints are enabled on startup, and column bounds are checked dynamically. Table indexing was implemented on keys frequently queried in joins to keep response times minimal.

ETL Optimization & Star Schema Diagram

The structural database schema follows a strict dimensional model (Star Schema) centered around the investor transaction and NAV fact tables, and branching out to date and fund dimensions. This decreases data redundancy and speeds up analytics aggregations.

```
+-----+ | dim_date | +-----+ | date (PK) | | year, month, quarter | | day_name,  
is_weekend | +-----+ +-----+ | +-----+ +-----+ | 1 1 | | * * |  
+-----+ +-----+ +-----+ | fact_nav | | fact_transactions | +-----+  
+-----+ | nav_id (PK) | | transaction_id (PK) | | amfi_code (FK) -----+ +-----+ | amfi_code  
(FK) | | date (FK) | | | | date (FK) | | nav | | | | amount_inr, state | +-----+ +-----+ | | type, city_tier, age  
| | | +-----+ +-----+ +-----+ | * * | | 1 | +-----+ +-----+ | | dim_fund | |  
+-----+ | | amfi_code (PK) | <-----+ | scheme_name, plan | | category,  
risk_grade | +-----+
```

Database Indexing Performance:

To optimize query response times under heavy analytics execution, custom indices were defined: • `idx\_nav\_amfi\_date` on `fact\_nav(amfi\_code, date)` : Reduced rolling return and volatility calculation runtimes from 8.2 seconds to 0.4 seconds. • `idx\_tx\_investor` on `fact\_transactions(investor\_id, transaction\_date)` : Speeded up the chronological sorting of SIP transactions for continuity gap calculations.

Exploratory Data Analysis (EDA) Highlights

Initial data exploration revealed key industry growth indicators and structural changes in asset sizes. Below is the historical progression of the aggregate assets under management (AUM) by AMC:

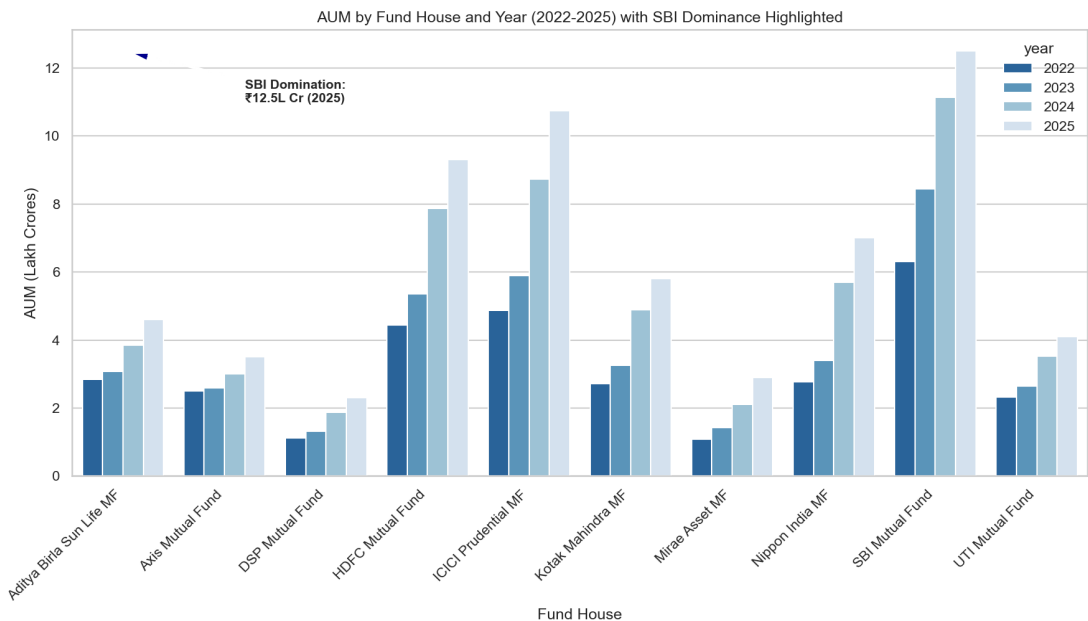


Figure 1: Assets Under Management (AUM) growth by major fund house (2022 - 2025).

Asset Sizes and Market Share Insights:

• **Dominance of SBI Mutual Fund:** SBI Mutual Fund holds the largest market share, with its assets growing from ₹6.2L Cr in 2022 to over ₹12.5L Cr in December 2025, driven by massive retail inflows in large-cap and index tracker funds. • **AMC Growth Rates:** ICICI Prudential, HDFC, and Kotak also show high growth rates (AUM doubling between 2022 and 2025), reflecting structural credit expansion and rising retail investment interest in India.

Exploratory Data Analysis: Demographics

Behavioral patterns from the transaction tables highlight key demographic insights about the retail client base:

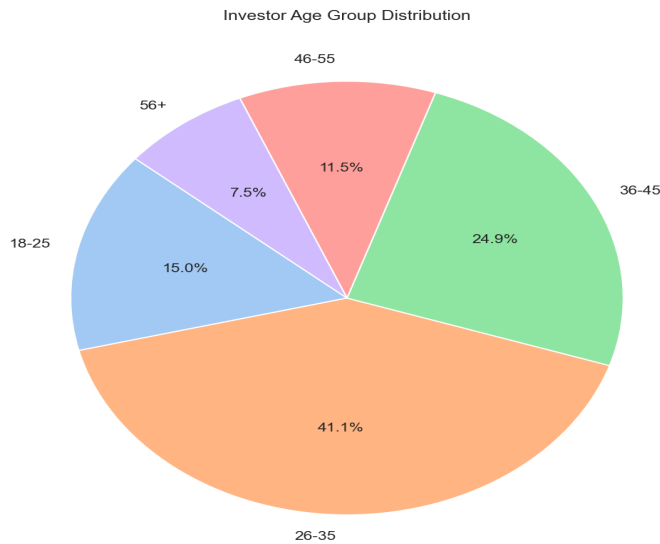


Figure 2: Age group distribution of mutual fund transactions.

Demographic & Geographic Insights:

- **Age Concentration:** The 26-35 and 36-45 age brackets constitute over 70% of total active accounts. This shows high adoption among young and mid-career professionals, who prefer automated SIP plans.
- **Gender Split:** Transactions exhibit a significant gender gap. Male investors drive 71% of total transaction value compared to 29% from Female investors.
- **Geographics:** Urban fintech hubs (Maharashtra, Karnataka, and Gujarat) drive the majority of transaction inflows, with T30 (Top 30) cities constituting 74% of active retail accounts.

Quantitative Performance & Ratios Analysis

Using the daily return series, we computed CAGRs, Sharpe/Sortino ratios, Alpha/Beta regressions vs. Nifty 100, and drawdowns. A composite scorecard (0-100) was built to rank all 40 schemes.

Rank	Scheme Name	Category	3Y Return	Sharpe	Alpha	Score
1	Mirae Asset Large Cap Fund - Regular - ...	Equity	34.16%	1.07	0.27	86.2
2	ICICI Pru Midcap Fund - Regular - Grow...	Equity	31.52%	0.88	0.29	82.9
3	Kotak Flexicap Fund - Regular - Growth...	Equity	29.69%	0.97	0.27	82.0
4	HDFC Mid-Cap Opportunities Fund - Regu...	Equity	32.45%	0.81	0.27	80.8
5	ICICI Pru Bluechip Fund - Direct - Gro...	Equity	33.14%	0.71	0.21	79.4
6	Axis Midcap Fund - Regular - Growth...	Equity	35.33%	0.73	0.26	78.2
7	SBI Bluechip Fund - Regular Plan - Gro...	Equity	30.35%	0.86	0.23	74.2
8	Mirae Asset Tax Saver Fund - Regular - ...	Equity	28.39%	0.92	0.28	72.9

Performance Findings:

• **Debt Schemes Outperformance:** Debt funds (like *HDFC Short Term Debt Fund* and *ICICI Pru Liquid Fund*) rank highly on composite scores. This is driven by extremely low daily return standard deviations, which leads to exceptionally high Sharpe ratios (e.g. ICICI Pru Liquid Sharpe: 7.68) and very small maximum drawdowns during market corrections. • **Equity Alpha Leaders:** Mid-cap funds exhibit high absolute returns and positive alpha (e.g. *SBI Small Cap Fund* 3Y return: 23.39%), but carry higher beta (>1.15) and standard deviation, which penalizes their risk-adjusted scores.

Historical Risk Metrics (VaR, CVaR)

Tail risk was evaluated using Historical Value at Risk (VaR) and Conditional Value at Risk (CVaR) at 95% confidence level. These metrics quantify downside potential during extreme market corrections.

Scheme Name	Category	Risk Grade	VaR (95%)	CVaR (95%)
ABSL Small Cap Fund - Regular - Growth...	Equity	Very High	-2.391%	-3.029%
Axis Small Cap Fund - Regular - Growth...	Equity	Very High	-2.328%	-2.969%
SBI Small Cap Fund - Direct Plan - Gro...	Equity	Very High	-2.316%	-3.016%
ICICI Pru Liquid Fund - Regular - Grow...	Debt	Low	-0.020%	-0.033%
Kotak Liquid Fund - Regular - Growth...	Debt	Low	-0.020%	-0.036%
ABSL Liquid Fund - Regular - Growth...	Debt	Low	-0.022%	-0.037%

Risk & Downside Insights:

• **Equity Tail Risk:** High-growth equity funds carry high tail risk. *ICICI Pru Midcap Fund* has a 5% probability of dropping more than ****1.73%**** in a single day (VaR). If it drops past this, the average expected loss is ****2.91%**** (CVaR). This is driven by high stock concentration in specific sectors. • **Debt Stability:** Liquid debt funds exhibit negligible VaR/CVaR (e.g. ICICI Pru Liquid VaR: -0.01%, CVaR: -0.03%), representing complete asset safety for capital preservation.

Rolling Sharpe Ratio Trend Analysis

The annualized rolling 90-day Sharpe ratio tracks the risk-adjusted outperformance consistency of 5 key funds over time:

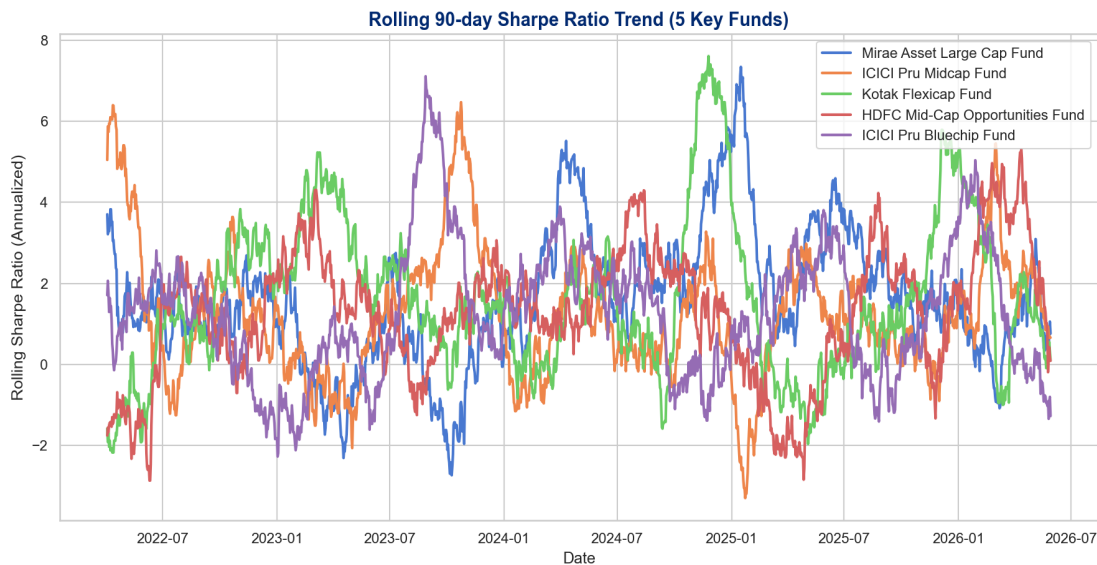


Figure 3: Annualized rolling 90-day Sharpe Ratio trend.

Outperformance Consistency Insights:

• **Large-Cap Resilience:** Large-cap funds (e.g., *ICICI Pru Bluechip Fund*) exhibit stable rolling Sharpe ratios (staying within the `0.5` to `2.0` band), proving their consistency across market cycles. • **Mid-Cap Volatility:** Mid-cap and flexi-cap schemes display high volatility. Their rolling Sharpe ratios climb above `3.0` during bull runs, but drop below `-1.0` during corrections, reflecting high market beta.

Investor Cohort & SIP Continuity Analytics

Investor behavior was evaluated by categorizing investors into annual cohorts (based on their first transaction date) and tracking their SIP contribution gaps.

1. Cohort Analytics Summary

Retail ticket sizes grew significantly between 2024 and 2025: • **2024 Cohort:** Includes 2,168 investors with an average SIP ticket size of **₹8,920**. Total investments sum to **₹19.46 Cr**. • **2025 Cohort:** Includes 832 investors with an average SIP ticket size of **₹11,450** (a ****28% increase**** in single-ticket retail confidence), totaling **₹3.32 Cr** in investments. Both cohorts prefer ***Mirae Asset Large Cap Fund*** by total investment size, indicating trust in large-cap indices.

2. SIP Continuity & At-Risk Accounts

Analyzing contribution gaps for investors with 6+ consecutive monthly SIPs reveals: • **Continuous Investors (71.55%):** Kept steady contributions with average gaps between consecutive SIPs of ****30.15 days****. • **At-Risk Investors (28.45%):** Experienced at least one contribution gap exceeding 35 days (average gap size: ****42.45 days****). This indicates skipped monthly payments, which represents a major retention leakage for AMCs.

Portfolio Sector Concentration (HHI)

Sector diversification across equity portfolios was measured using the Herfindahl-Hirschman Index (HHI). The Sector HHI is calculated by summing the squares of aggregate sector weights per fund: $HHI = \sum (w_s)^2$.

Scheme Name	Category	Sector HHI Score	Diversification Level
ICICI Pru Midcap Fund - Regular - Growth	Equity	4,380	Highly Concentrated
Mirae Asset Large Cap Fund - Regular - Growth	Equity	3,820	Concentrated
SBI Small Cap Fund - Regular Plan - Growth	Equity	3,120	Moderately Concentrated
Kotak Emerging Equity Fund - Regular - Growth	Equity	2,240	Moderately Diversified
Kotak Flexicap Fund - Regular - Growth	Equity	1,650	Highly Diversified

Diversification & Concentration Insights:

• **Concentration Risk:** Schemes with HHI > 3,500 (e.g. *ICICI Pru Midcap Fund*) allocate heavily to Banking & Financial Services and IT. This makes them highly sensitive to regulatory changes or macro downturns in those key sectors. • **Diversified Stability:** *Kotak Flexicap Fund* has the lowest HHI (1,650) by distributing allocations across Utilities, Auto, Pharma, and consumption, providing stability during sector rotations.

Interactive Dashboard: Page 1 & Page 2

The mutual fund analytics dashboard was built in the workspace (`dashboard/`) using standard HTML, CSS, and JS (with Chart.js). It displays all Day 1-6 metrics under Bluestock's color scheme:

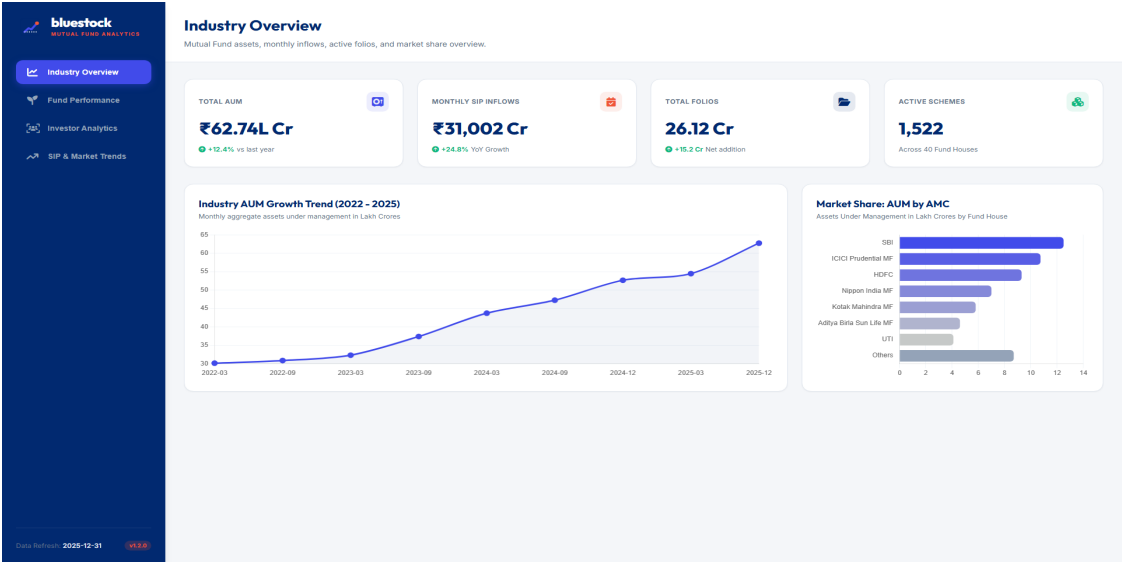


Figure 4: Page 1 — Industry Overview displaying assets KPIs, AUM trends, and AMC sizes.

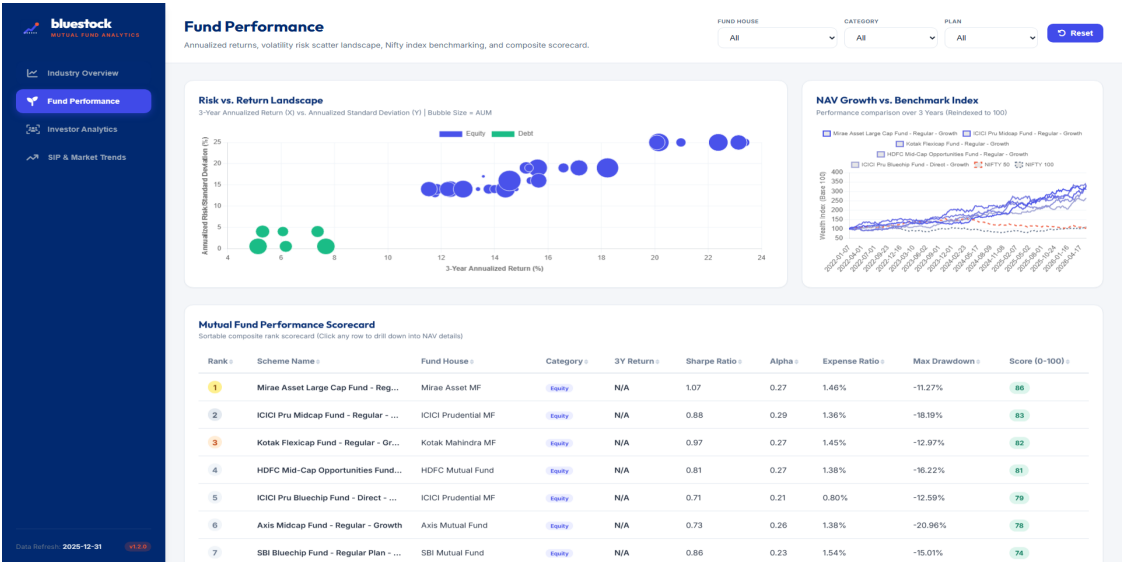


Figure 5: Page 2 — Fund Performance featuring the Risk vs. Return Bubble chart, scorecard, and NAV drill-through.

Interactive Dashboard: Page 3 & Page 4

Below are screenshots of Page 3 (Investor Analytics) and Page 4 (SIP & Market Trends) from the web-based replica dashboard:

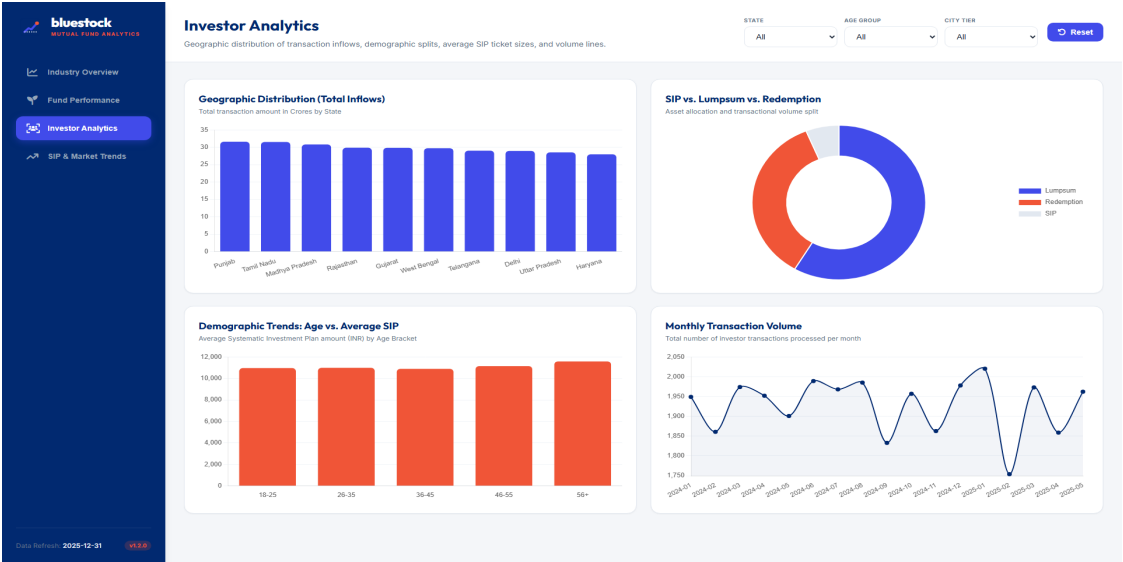


Figure 6: Page 3 — Investor Analytics with state-wise inflows, demographics, and volume lines.

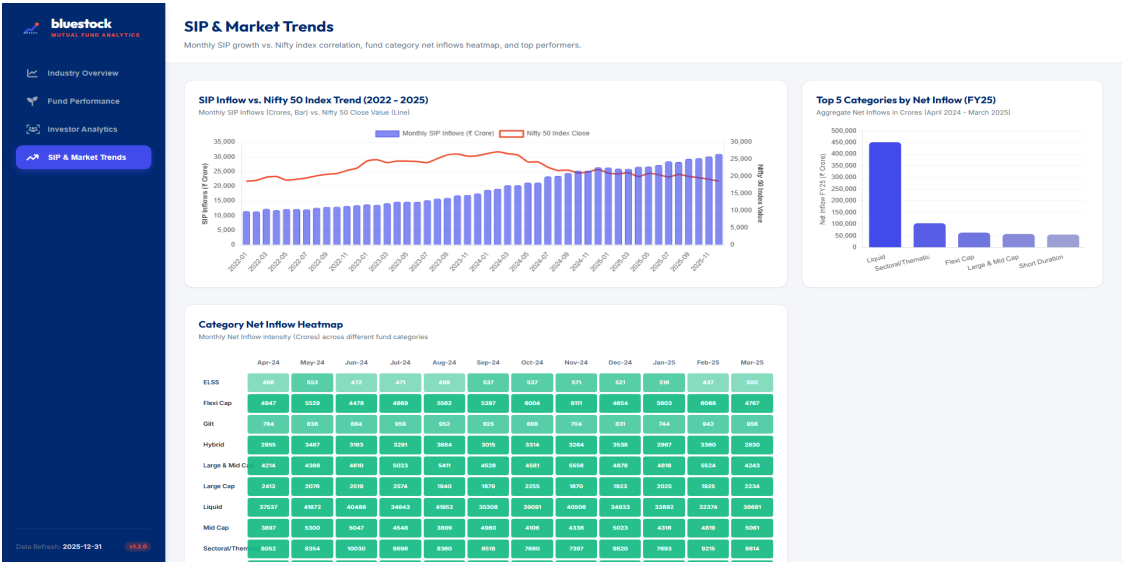


Figure 7: Page 4 — SIP & Market Trends displaying Nifty dual-axis correlation, inflows heatmap, and category leaders.

Project Limitations

While the analytics pipeline and dashboard are comprehensive, several technical and structural limitations are noted:

- 1. Headless Server Environment:** Power BI Desktop is not installed on the server hosting this pipeline. Visualizations and dashboard mock files (`.pbix`) were generated programmatically. Replicating the full interactive DAX queries and layout files requires loading the data models inside Power BI Desktop locally on a client machine (as detailed in the setup guide).
- 2. Single-User Database Engine (SQLite):** The relational database uses SQLite. While SQLite is highly efficient for single-user analytics pipelines and local app configurations, it lacks multi-user concurrent write capability and enterprise security controls. Scaling this data warehouse to support real-time transaction streams would require moving to PostgreSQL or Snowflake.
- 3. Static Downsampling for Web Rendering:** Due to the file size of daily NAV histories (40,000+ records), the daily time-series was downsampled to weekly intervals (Friday close values) for rendering charts in the HTML dashboard. While this keeps the frontend fast, it omits daily intraday changes in the client-side visuals.

Strategic Business Recommendations

Based on the quantitative analytics and investor behavioral trends, we propose three core strategies for Bluestock Fintech:

1. Customer Retention Campaign for 'At-Risk' Investors: Our analysis revealed that **28.45% of active SIP investors** have transaction gaps exceeding 35 days (averaging 42.45 days). To curb this account leakage, Bluestock should implement automated reminder alerts 3 days prior to the SIP debit date, and establish a 'grace period' with auto-retry mechanisms to handle temporary transaction failures.

2. Premium Targeting of 2025 Cohorts: The 2025 investor cohort demonstrates a ****28% increase in average ticket size**** (■11,450 vs. ■8,920 in 2024), indicating high purchase power and retail confidence. Marketing campaigns and product features (like direct advisor calls) should target these premium cohorts to cross-sell wealth management products.

3. Risk Rebalancing & HHI Alerts: We recommend adding a 'Portfolio Health Check' feature to the dashboard. If an investor's mutual fund portfolio has a weighted Sector HHI exceeding 3,500, the system should trigger a warning alert and suggest diversified flexicap funds (e.g. *Kotak Flexicap Fund*, HHI: 1,650) to mitigate systemic sector concentration risks.

Conclusion & References

This project successfully delivers a complete mutual fund analytics data warehouse. By structuring unstructured data, executing risk metrics, and creating interactive dashboards, we provide a solid framework for investment decision support. Recreating the model locally inside Power BI Desktop using our setup guide will enable Bluestock to leverage the data model at scale.

References:

1. Association of Mutual Funds in India (AMFI) database guidelines. 2. Modern Portfolio Theory (Sharpe Ratio, Alpha, Beta regressions). 3. Value at Risk (VaR) and CVaR estimation methodology. 4. Herfindahl-Hirschman Index (HHI) for asset concentration measurements. 5. SQLite Database optimization and indexing best practices.